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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
IV Semester B. E. Examinations October-2023
Electronics and Communication Engineering
FIELDS AND WAVES

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

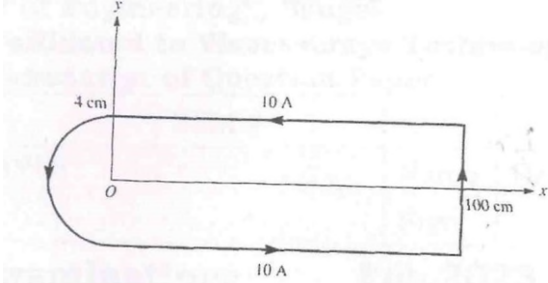
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	State Coulomb's Law. Write an expression for Electric field due a unit charge at an arbitrary point $P(x, y, z)$.	02
	1.2	A charge Q_0 located at the origin in free space produces a field for which $E_z = 1kV/m$ at point $P(-2,1,-1)$. Find Q_0 .	02
	1.3	Uniform line cahrgs of $0.4\mu C/m$ and $-0.4\mu C/m$ are locaed in the $x = 0$ plane at $y = -0.6$ and $y = 0.6m$ respectively. Find E at $Q(2,3,4)$.	02
	1.4	The potential is given by $V = 100 + 50\rho + 150\rho\sin\phi V$. Find E at $P(1,60^\circ, 0.5)$	02
	1.5	Given the field $H = 20\rho^2\hat{\phi} A/m$. Find the total current.	02
	1.6	Planar current sheets of $K = 30\hat{a}_z A/m$ and $-30\hat{a}_z$ are located in free space at $x = 0.2$ and $x = -0.2$ respectively. For the region $-0.2 < x < 0.2$. Find H .	02
	1.7	Give four analogies between Electrostatics and Magneto statics.	02
	1.8	With releveant expressions explain Scalar Potential and Vector Potential.	02
	1.9	State Maxwell's equation for time varying field in Point form.	02
	1.10	Most microwave ovens operate at $2.45GHz$. Assume that $\sigma = 1.2 \times 10^6 S/m$ and $\mu_R = 500$ for the stainless-steel interior, and find the depth of penetration .	02

PART B

2	a	A uniform line charge densiy $\rho_L n C/m$ of Length 'L' is placed along z-axis.	
		i) Derive an expression for elctric field intensity E at an arbitrary point $P(x, y, z)$.	
		ii) If the line charge density $\rho_L n C/m$ is of infinite length then, determine E at any point (x_1, y_1, z_1) using the result i.	08
	b	Given $\rho_v = 5n C/m^3$ for $0 < \rho < 1mm$ and no othern charges are present. Find	
		i) Total flux pasing through surface $0.5 < \rho < 1.5mm$, $0 < z < 1m$.	
		ii) D_p for $\rho < 1mm$	
		iii) D_p for $\rho > 1mm$	08

3	a	A volume charge distribution with spherical symmetry is $\rho_v = \begin{cases} \rho_0 & 0 \leq r \leq R \\ 0 & 0 \geq R \end{cases}$ i) Find E and V for $r > a$ ii) Find E and V for $r < a$	10
	b	Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0, \epsilon_{r1} = 4$ for $z \leq 0, \epsilon_{r2} = 3$. A uniform electric field $E_1 = 5\hat{x} - 2\hat{y} + 3\hat{z} \text{ K V/m}$ exists for $z \geq 0$. Find: i) E_2 for $z \geq 0$ ii) The angles E_1 and E_2 make with the interface. iii) The energy densities in J/m^3 in both dielectrics.	06
OR			
4	a	Given a surface charge density of 8 nC/m^2 on the plane $x = 2$, a line charge density of 30 nC/m on the line $x = 1, y = 2$, and a $1 \mu\text{C}$ point charge at $p(-1, -1, 2)$. Find V_{AB} for points $A(3, 4, 0)$ and $B(4, 0, 1)$.	08
	b	The region between concentric spherical conducting shells $r = 0.5 \text{ m}$ and $r = 1 \text{ m}$ is charge free. If $V(r = 0.5) = -50 \text{ v}$ and $V(r = 1) = 50 \text{ V}$. Using Laplace equation, i) Dervie the expression for V, E and capacitance C . ii) Compute the V, E and C iii) Determine the potential distribution and the electric field strenght in the region between the shells.	08
5	a	A solenoid of length ' l ' and radius ' r ' consists of N turns of wire carrying current I . i) Derive an expression for Magnetic field intensity H at center of Solenoid. ii) Dervie an expression for Magnetic field intenstiy H at end of Solenoid.	08
	b	An infinite filament on the z axis carries $20 \pi \text{ mA}$ in the $+z$ direction. Three uniform cylindrical current sheets are also present 400 mA/m at $\rho = 1 \text{ cm}$, -250 mA/m at $\rho = 2 \text{ cm}$, and -300 mA/m at $\rho = 3 \text{ cm}$. Calculate H at $\rho = 0.5, 1.5, 2.5$, and 3.5 cm .	08
OR			
6	a	For the filamentary loop shown in fig 6a, find the Magnetic Field Intensity at O . 	08
	b	An infinitely long coaxial transmission line along z axis has inner radius ' a ' and carries current ' I ' while the outer conductor has radius ' b ' with thickness ' t ' and carries a current $-I$. Determine $\vec{H}$ at a radial distance ' ρ ' such that i) $\rho < a$ ii) $a < \rho < b$ iii) $b < \rho < b + t$ iv) $\rho > b + t$	08

7	a	An interface of two magnetic materials has a linear current density $K \text{ A/m}$. Derive a relation between i) The tangential components of magnetic field intensities in two regions. ii) The normal components of magnetic flux density at the interface.	08
	b	i) Derive equation of continuity ii) Using the expression derived in (i), show that Ampere circuital law cannot be applicable for time varying currents. iii) Modify Ampere circuital law such that it is applicable for time varying currents.	08
8	a	State the Poynting theorem and derive the expression to prove the Poynting theorem.	06
	b	A uniform wave in air has, $E = 10(\cos 2\pi \times 10^6 t - \beta z)\hat{y} \text{ V/m}$ Find: i) β ii) λ iii) The intrinsic impedance iv) Electric Field	04
	c	In free space ($z \leq 0$), a plane wave with $H = 10\cos(10^8 t - \beta z)\hat{x} \text{ A/m}$ is incident normally on a lossless medium $\epsilon = 2\epsilon_0$ and $\mu = 8\mu_0$ in region $z \geq 0$. Determine: i) Reflection coefficient and transmission coefficient ii) The Electric and Magnetic transmitted wave iii) The Electric and Magnetic reflected wave	06